Mechanical Prototyping

Activity 1

Paper Prototyping for Problem Identification

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Batch: A2

Project Title: *Twist & Tidy – A Finger-Operated Cable Organizer*

Problem Statement

Cables—chargers, earphones, USBs—often end up tangled, messy, or damaged. Most existing organizers are bulky, hard to use, or don't support different wire sizes. Especially in student setups or rural homes, there's a need for a simple, reusable, and low-cost solution to keep wires neat and accessible.

Aim of the Idea

To design a compact, twistable cable organizer that allows users to wind and store wires using just their fingers—no tools, no hassle.

Objectives

- Design a modular clip that can wind and hold cables of various sizes (3–8 mm).
- Enable tool-free assembly using snap-fit parts.
- Make the product reusable, durable, and easy to manufacture.
- Ensure the design is suitable for 3D printing and rural accessibility.

Output / Purpose

A working prototype of a cable organizer that:

- Neatly winds and stores cables.
- Prevents tangling and damage.
- Can be reused multiple times.
- Is lightweight, portable, and easy to mount or carry.
- Can be manufactured affordably for student and community use.

Working Principle

- Place the wire into the curved grooves of the center part.
- Snap on the top and bottom shells.
- Twist the clip using your fingers.
- The wire coils neatly inside the grooves.
- Unclip to remove or reconfigure the wire.

Key Parts of the Design:

- **Top Shell**: This is the part you hold and twist with your fingers. It locks onto the center and bottom parts, helping you rotate the whole clip to wind the wire.
- **Center Grip Core**: This middle piece has wavy grooves or curved channels. You place the wire into these grooves, and they guide the wire as it coils during rotation. It's the heart of the winding mechanism.
- **Bottom Shell**: Similar to the top shell, it completes the enclosure and provides grip from the other side. Together with the top shell, it holds the center part securely.
- **Cable Grooves**: These are the curved paths inside the center grip core where the wire sits. As you twist the clip, the wire follows these grooves and winds up neatly.

Drawing:

